[Def. A Complete Normed Space X is called a Banach Space.

[Thm. Let X be a Normed Space. Then, the followings are equivalent:
1). X is Complete.
2). If a sequence $\{x_n\}$ in X satisfies $\sum_{n=1}^{\infty}\|x_n\| < \infty$, then $\sum_{n=1}^{\infty} x_n$ converges.
Proof.